Examiner only

8. (a) According to the Journal of the British Dental Association, the increase in consumption of fruit juices and other acidic drinks is believed to be one of the leading causes of dental erosion in children and adolescents.

Many fruit juice drinks contain more than one type of acid. Citric acid can be used as a natural preservative and provides a sour taste. Ascorbic acid (vitamin C) is a water-soluble vitamin that must be consumed regularly to ensure proper body function. Citrus fruits, as well as tomatoes and other fresh vegetables, are good sources of vitamin C.

The table below shows information about the content of different fruit juice drinks.

Fruit juice drink	pH	Water (%)	Citric acid (%)	Ascorbic acid (mg / 100 g)	Sugar (%)
lime	2.2	76.4	4.80	29	1.68
lemon	2.4	81.3	4.60	53	1.80
grapefruit	3.0	90.3	1.35	38	2.34
orange	3.6	87.4	0.96	50	4.85
tangerine	3.8	85.7	0.74	31	7.89

(i) Tick (✓) the **two** conclusions that can be drawn from the information. [2]

as pH increases, citric acid content decreases and sugar content increases	
as acidity decreases, ascorbic acid content decreases and water content decreases	
tomatoes are a good source of vitamin C and citric acid	
citrus fruits contain ascorbic acid and a natural preservative	

(ii) **Use information from the table** to suggest why citric acid content has a much greater effect on the pH than ascorbic acid content. [1]

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(b) The concentration of a solution of sodium hydroxide can be determined by titrating with sulfuric acid. The equation for the reaction is shown.

$$H_2SO_4 \quad + \quad 2NaOH \longrightarrow Na_2SO_4 \quad + \quad 2H_2O$$

(i) Give the **ionic** equation for the formation of water in any neutralisation reaction. Include state symbols. [2]

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(ii) 21.0 cm^3 of sulfuric acid with a concentration of 0.350 mol / dm^3 neutralised 25.0 cm^3 of the sodium hydroxide solution.

I. Calculate the number of moles of sulfuric acid used in the reaction. [1]

Number of moles = ... mol

II. Calculate the concentration of the sodium hydroxide solution. [2]

Concentration = ... mol / dm^3

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(iii) During a similar titration reaction, 0.36 g of water was produced. Calculate the number of **molecules** of water produced in this reaction.

Give your answer in **standard form**. [2]

$A_r(H) = 1$ $A_r(O) = 16$

Avogadro's constant = 6.0×10^{23}

Number of molecules = ...

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